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17 March 2026

Methods of Planning Analysis

Prof. Powers

Assignment 2: Survey

For my survey, I am investigating a research question about the MTA's recent subway route change of the F and M trains (F/M Swap). My question is "Has the F/M Swap impacted the commute of passengers who originate or terminate trips at the affected stations?" My target population for this survey is only people who take the affected line (the F and M trains), and only people who begin or end their journey at either 57 St, Lexington Av / 63 St, Roosevelt Island, and 21 St – Queensbridge.

To carry out this survey, I placed QR code survey flyers (see below) in the mezzanine and platform areas of the four affected stations. Ideally, to accurately find a representative sample, I would gain access to the database of OMNY card taps at these four stations and send survey request emails to a random sample of these OMNY card users. Doing this would ensure even distribution across the stations, as well as feedback from those who use the service at different times in the day. With a 95% confidence interval and a 10% Margin of Error, my ideal sample size to make statistically significant observations for this survey would be 100 respondents. In terms of data stratification, the only groups I would form would be per-station, though I anticipate that the results are consistent across all stations.

In terms of the survey itself, I've designed a ten-question survey that contains nominal questions about personal ridership behavior, ordinal closed-ended questions about changes due to the F/M swap, ratio questions about minutes added or saved due to the swap, and lastly and open-ended question meant to capture general sentiment. When crafting this survey, I assumed that most responses would say that their trip times have increased. For that reason, I have added both ordinal and ratio-typed questions to understand the frequency and magnitude of this complaint. Additionally, I have added questions to understand if riders have been forced to change their behaviors, if they agree with the MTA's decision-making, and if they were informed before the change was made.

With this survey, I am most concerned with coverage and non-response errors. Given the relative recency of the change, as well as the potential for disruption for specific commuters, my method of posting QR codes in the station may have introduced a bias where commuters who have been impacted more severely may have been more inclined to fill the survey out than those with minimal impacts. Additionally, I was originally concerned about a sampling error, though I was pleasantly surprised to find that I had received 150 responses, which exceeded the 100 that I needed to make a statistically significant observation.

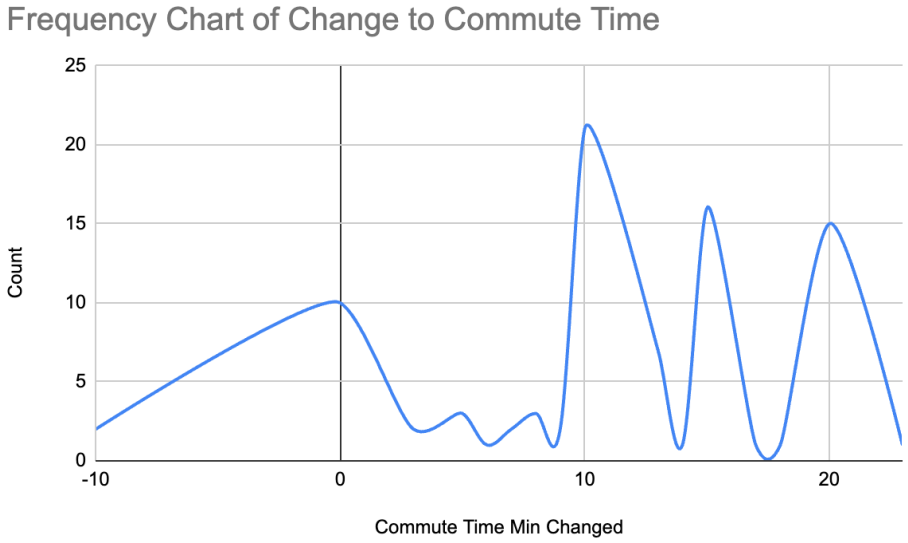
From my data, I created variables such as **change to commute time in minutes**, **opinion on MTA's decision to swap the trains,** and **change to commuting behavior.** In the tables and charts below, you can see the results. If given more time and resources, I would investigate the relationship between station usage, trip frequency, and trip role in the values of these individual variables.

Survey Results

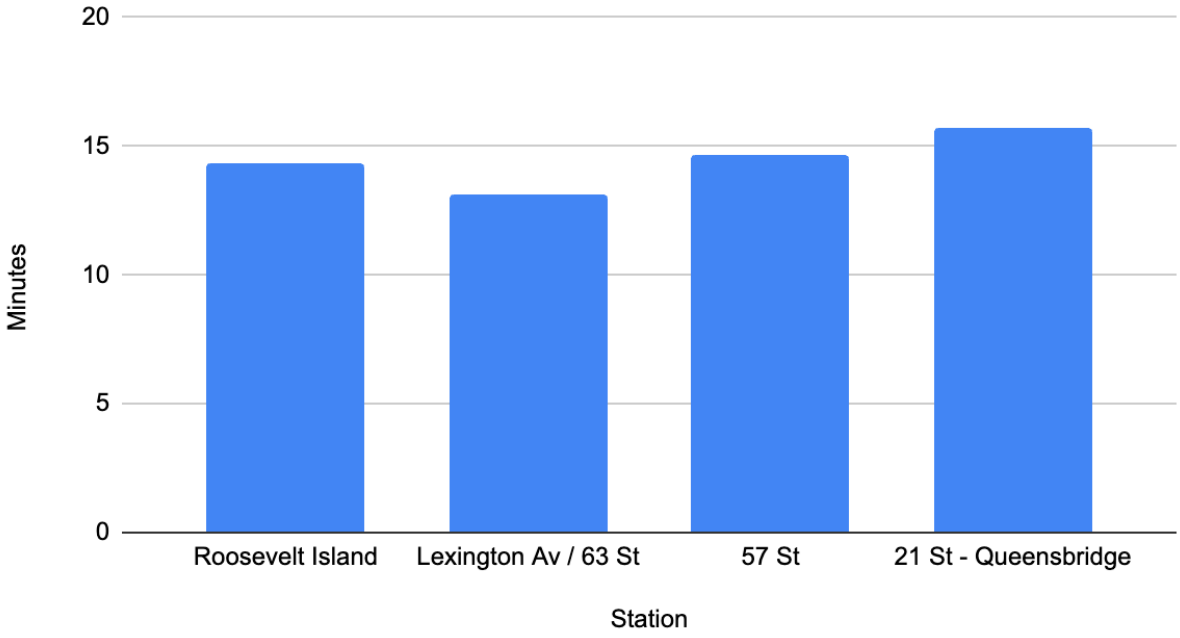
Question	Average Response
Commute time change	Added an average of 12 minutes Std. Dev 8 minutes 86% of respondents' trips got longer
Did MTA make the correct decision	Average in between Disagree and Strongly Disagree, with Strongly Disagree being the Mode
Informed beforehand	91% of commuters knew beforehand
Type of change to commute	20% unchanged 10% sometimes changed due to route optionality 50% changed their departure times 20% changed the train or route that they take

Analysis of open-ended question results (Total: 108 responses)

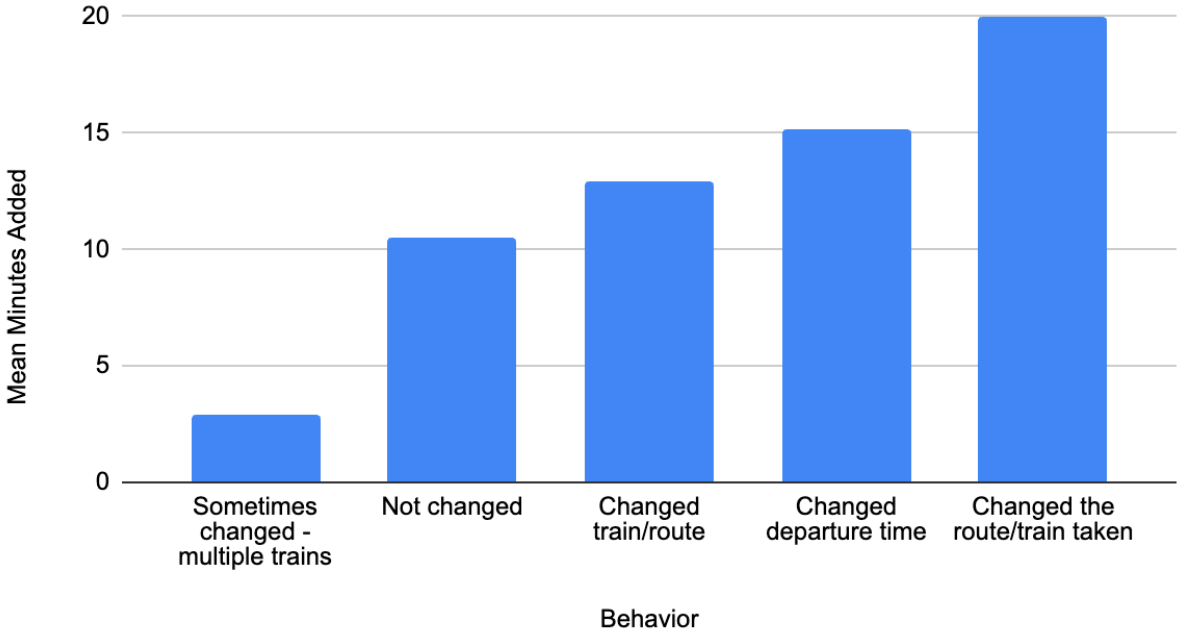
Open Ended Question Sentiment	% of comments mentioning
Responses complain about the train frequency of the M train. Specifically, they note that frequencies do not match the MTA's promises	35%
The M train is 8 cars compared to the F train's 10 cars, and therefore trains are more crowded, and passengers have to run down the platform.	19%
Roosevelt Island riders have no other alternative subway, and feel stranded by tourist-tram and infrequent ferry.	21%
The M train is local in Queens, taking longer to reach Eastern Queens.	16%
Riders are confused by the change from the M train to the F train on nights after 9:30 PM and on weekends.	9%
Riders like the new service to Astoria, but think the MTA needs to add more trains.	8%



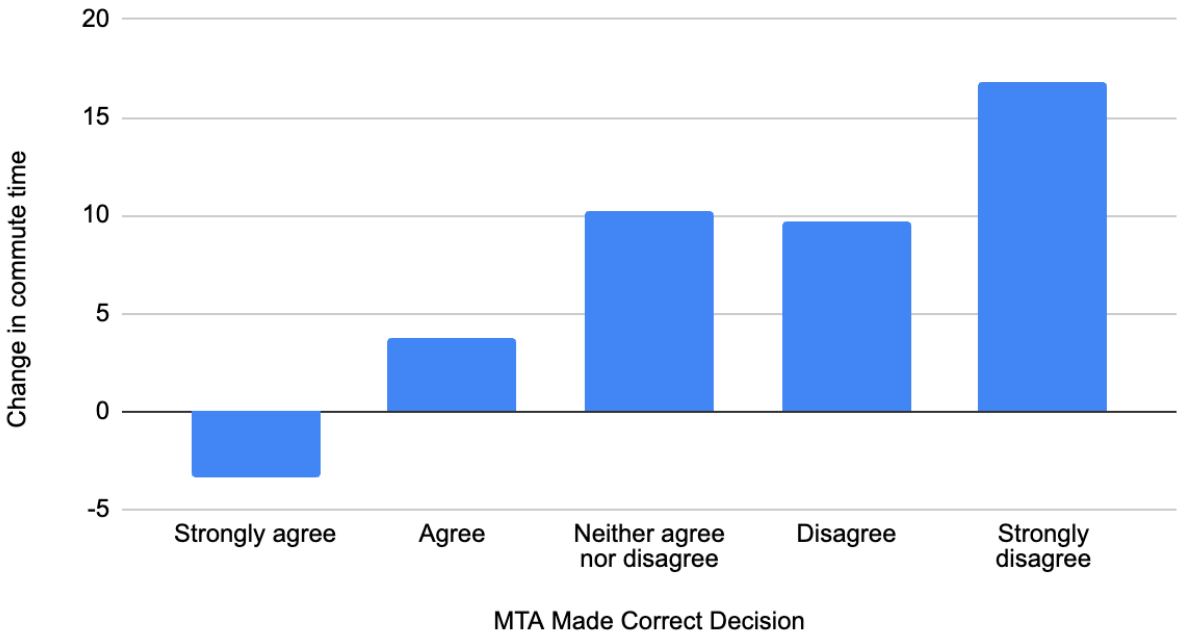
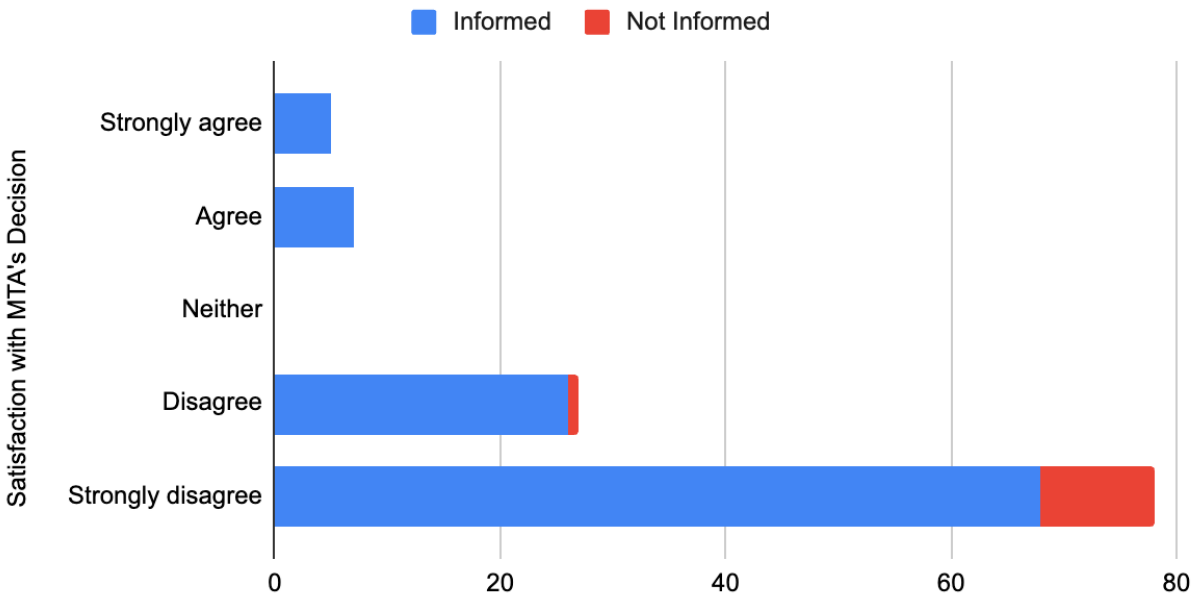
Minutes Added to Commute By Station



Minutes added to commute compared to behavior



Change to commute time vs. satisfaction with MTA's decision



Satisfaction with MTA's decision compared to whether riders were informed of the change beforehand